Conditional probability, Bayes, binomial CHEAT SHEET

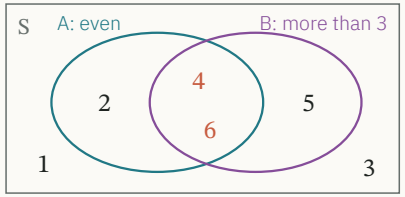
PROBABILITY: THE BASICS

S = sample space: every outcome one die: S = {1, 2, 3, 4, 5, 6}

event: a part of S even = {2, 4, 6}

$P(E) = \frac{\text{outcomes in E}}{\text{outcomes in S}}$ (equally likely) $P(\text{even}) = \frac{3}{6} = \frac{1}{2}$

$0 \le P(E) \le 1$ $P(S) = 1$ $P(\varnothing) = 0$



$A \cap B = \{4, 6\}$

$A \cup B = \{2, 4, 5, 6\}$

overlap counted twice: take it away once

CONDITIONAL PROBABILITY

$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$ $P(B) \neq 0$

six given even: $\frac{1/6}{3/6} = \frac{1}{3}$ even given six: 1

$P(A \mid B) \neq P(B \mid A)$ in general

$P(\text{not A} \mid F) = 1 - P(A \mid F)$ $P(S \mid F) = 1$

~~1~~ 2 ~~3~~ 4 ~~5~~ 6 given even: the world shrinks

	won	not won	total
rain	3	7	10
dry	18	12	30

$P(\text{win} \mid \text{rain}) = 3/10$ $P(\text{win} \mid \text{dry}) = 18/30 = 3/5$

INDEPENDENT EVENTS

independent: $P(A \cap B) = P(A)\,P(B)$ $(P(A \mid B) = P(A))$

one die: even and ≤ 2: $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$ independent

even and > 3: $\frac{1}{3} \neq \frac{1}{4}$ dependent

heart, king: $\frac{1}{4} \times \frac{1}{13} = \frac{1}{52}$ independent

at least one = 1 − P(none): two alarms $1 - 0.1 \times 0.1 = 0.99$

three dry days at 0.7: $0.7^3 = 0.343$

independent ≠ mutually exclusive

exclusive events (each P > 0) are always dependent

NOT, OR, THE ADDITION RULE

$P(\text{not A}) = 1 - P(A)$ $P(\text{not a six}) = \frac{5}{6}$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

even or more than 3: $\frac{3}{6} + \frac{3}{6} - \frac{2}{6} = \frac{2}{3}$

mutually exclusive (no overlap): $P(A \cup B) = P(A) + P(B)$

heart or king: $\frac{13 + 4 - 1}{52} = \frac{4}{13}$

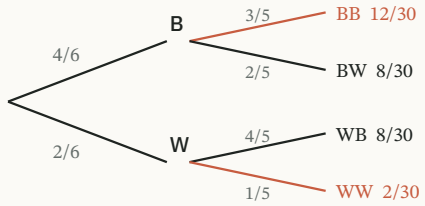
at least one six in two dice: $1 - \frac{25}{36} = \frac{11}{36}$

MULTIPLICATION RULE AND TREES

$P(A \cap B) = P(A)\,P(B \mid A) = P(B)\,P(A \mid B)$

two kings, no replacement: $\frac{4}{52} \times \frac{3}{51} = \frac{1}{221}$

along a path: multiply across paths: add



matching pair: 12/30 + 2/30 = 7/15

TOTAL PROBABILITY AND BAYES' THEOREM

partition $E_1, \ldots, E_n$: no overlaps, together all of S

total: $P(A) = \Sigma\, P(E_i)\,P(A \mid E_i)$

Bayes: $P(E_i \mid A) = \frac{P(E_i)\,P(A \mid E_i)}{\Sigma\, P(E_j)\,P(A \mid E_j)}$

faulty bolt: 0.010 + 0.009 + 0.010 = 0.029

$P(\text{C} \mid \text{faulty}) = \frac{0.010}{0.029} = \frac{10}{29} \approx 0.34$

prior $P(E_i)$ → evidence A → posterior $P(E_i \mid A)$

THE MEDICAL TEST, AS PEOPLE

1 in 1,000 ill · catches 99% · false alarm 1%

100,000 people: 100 ill → 99 test +

99,900 well → 999 test +

$P(\text{ill} \mid +) = \frac{99}{99 + 999} = \frac{11}{122} \approx 0.09$

a second positive test: ≈ 0.91

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RANDOM VARIABLES AND DISTRIBUTIONS

x	0	1	2
P(X = x)	1/4	1/2	1/4

X = heads in two coins

random variable X: a number for every outcome

$0 \le p_i \le 1 \quad \Sigma\, p_i = 1$

find k: $P(X = x) = kx$, $x = 1, 2, 3, 4 \rightarrow 10k = 1$, $k = \frac{1}{10}$

two cards, X = kings: $\frac{188}{221}$, $\frac{32}{221}$, $\frac{1}{221}$

MEAN, VARIANCE, STANDARD DEVIATION

$E(X) = \mu = \Sigma\, x_i\, p_i$ (the balance point)

$Var(X) = \Sigma\, (x_i - \mu)^2\, p_i = E(X^2) - [E(X)]^2$

$\sigma = \sqrt{Var(X)}$

die: $E(X) = \frac{21}{6} = 3.5 \quad E(X^2) = \frac{91}{6}$

$Var(X) = \frac{91}{6} - 3.5^2 = \frac{35}{12} \approx 2.92 \quad \sigma \approx 1.71$

fair game: expected gain = 0 raffle: $2.5 - 5 = -2.5$

BERNOULLI TRIALS

1. a fixed number of trials, n
2. the trials are independent
3. two outcomes: success, failure
4. the same p every time; $q = 1 - p$

THE BINOMIAL DISTRIBUTION

$X \sim B(n, p)$: $P(X = r) = \binom{n}{r} p^r\, q^{n-r}$

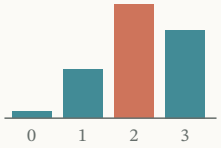
$\binom{n}{r} = \frac{n!}{r!\,(n-r)!} \quad \binom{3}{2} = \frac{3!}{2!\,1!} = 3$

the terms of $(q + p)^n$: each path has r successes

Galton board, 10 rows: middle bin $\frac{252}{1024} \approx 0.25$

3 free throws at 0.7:

P(0) = 0.027
P(1) = 0.189
P(2) = 0.441
P(3) = 0.343



p = 0.2 p = 0.5 p = 0.7

n = 10: the peak moves with p

ITS MEAN AND VARIANCE

$E(X) = np \quad Var(X) = npq \quad \sigma = \sqrt{(npq)}$

10 free throws at 0.7: mean 7, variance 2.1, $\sigma \approx 1.45$

3 coins: mean 1.5, variance 0.75

WORKED PROBLEMS

quiz, 10 guesses at ¼: $P(X = 3) = 120 \times \frac{1}{64} \times (¾)^7 \approx 0.25$

$P(\text{at least one right}) = 1 - (¾)^{10} \approx 0.944$

chargers, 5% faulty, box of 20: $P(\text{none}) = 0.95^{20} \approx 0.36$

$P(\text{at most one}) \approx 0.36 + 0.38 = 0.74$

$P(\text{at least one head}) > 0.99$: $2^n > 100 \rightarrow n = 7$

spam: $\frac{0.3 \times 0.4}{0.3 \times 0.4 + 0.7 \times 0.02} = \frac{60}{67} \approx 0.90$

WHICH RULE?

both happen (and)	multiply: P(A) P(B | A)
one or the other (or)	add, minus the overlap
at least one	1 − P(none)
you know B happened	shrink to B: P(A ∩ B) / P(B)
does B change A?	test P(A ∩ B) = P(A) P(B)
one of several causes	total probability: add the paths
the cause, given evidence	Bayes: one path / all the paths
successes in n trials	binomial, mean np